Current-source requalification: TIRust pin `9a781998`

Rehearsal on an unmerged pin. These pages qualify MadGraph `624ef69f05` with TIRust `9a781998e54a39fb06d3621de3cd458da09d2a52`, the head of TIRust PR 35 (not yet on TIRust `main`). The final pin must be a merged `main` commit; the same commands are rerun on it. They supersede the earlier pages only for this source and runtime (DSO `6c11cfd9eecf`). Shared-host results; precise paired timing acceptance remains Hua-Sheng's local TODO.

The fresh `install tirust` build of this pin passes `smoke --require-poles`, the pinned upstream C-ABI fixtures (27 DP and 54 QP lanes) and links only `libgcc_s`, `libm` and `libc`. Exports were regenerated from the campaign's frozen cards, models and `-O3` flags; the baseline is the starting revision 9228fc58.

Ordinary matrix: 11 processes × 6 lanes × 100 points

Table 1: Born, finite and both poles against starting-revision CutTools (`LoopMEComparator`). 66 of 66 lanes pass `3e-6`; 54 meet the separate `1e-8` target. Misses count scalar comparisons (split-order rows duplicate). "Previous" is source 0fcd2cd873 / runtime 854b79dc; the earlier EW 2⟶4 TIRust lane (40 DP / 60 QP) was recorded separately.

Process	3e-6	1e-8 misses CT/Ninja/COLLIER/ NinjaRS/TIRust	TIRust max finite residual	TIRust DP/QP	previous DP/QP
QCD u ū → Z	6/6	0/0/0/0/0	5.4e-14	100/0	100/0
QCD g g → t t̄	6/6	0/0/0/0/0	2.8e-12	100/0	100/0
QCD u ū → Z g g	6/6	0/56/56/56/56	2.8e-6	100/0	87/13
QCD u ū → Z Z g g	6/6	0/88/80/80/78	4.1e-7	100/0	14/86
EW u d̄ → W⁺	6/6	0/0/0/0/0	9.4e-13	100/0	100/0
EW e⁺e⁻ → μ⁺μ⁻	6/6	0/0/0/0/0	2.3e-13	100/0	100/0
EW e⁺e⁻ → t t̄ γ	6/6	0/0/0/0/0	1.1e-9	100/0	100/0
EW e⁺e⁻ → μ⁺μ⁻ γ γ	6/6	0/12/10/14/8	1.1e-7	99/1	–
QCD+QED u ū → g γ	6/6	0/0/0/0/0	2.6e-11	100/0	100/0
loop-induced g g → H	6/6	0/0/0/0/0	1.5e-10	100/0	100/0
loop-induced g g → H H	6/6	0/0/0/0/0	6.4e-10	100/0	100/0

TIRust now accepts double precision where the previous runtime needed a genuine QP rescue: its internal double-double and MPFR tensor tiers recover refused lanes before MadLoop rebuilds a QP numerator. In QCD $u\bar{u} \to Zgg$, Ninja, COLLIER, NinjaRS and TIRust miss the `1e-8` target on the same 28 points with the same residual, which points at the CutTools reference; the comparison itself assigns no independent truth.

Double-double gate (headroom 26)

Table 2: Every bridge call of the TIRust lane was captured (`tools/tirust_capture`) and replayed: the production build reproduces all recorded bits; references with headroom 40 (the previous 2^−66 gate) and 105 (no lane in DD) evaluate the same lanes at 256 bits. Error = max |difference| / max |value| of the same tensor degree. Flag 1e-12, fail 1e-9. 6131 DD calls in total.

Process	DP calls	DD calls	new-window calls	max error new window	max error all DD	verdict
EW u d̄ → W⁺	1212	0	0	—	—	not exercised
EW e⁺e⁻ → μ⁺μ⁻	5252	0	0	—	—	not exercised
EW e⁺e⁻ → t t̄ γ	40400	0	0	—	—	not exercised
EW e⁺e⁻ → μ⁺μ⁻ γ γ	246036	1545	60	1.9e-24	1.9e-24	qualified
loop-induced g g → H	808	0	0	—	—	not exercised
loop-induced g g → H H	3232	0	0	—	—	not exercised
QCD+QED u ū → g γ	5656	0	0	—	—	not exercised
QCD u ū → Z	202	0	0	—	—	not exercised
QCD g g → t t̄	5656	0	0	—	—	not exercised
QCD u ū → Z g g	23432	134	8	1.9e-27	3.6e-24	qualified
QCD u ū → Z Z g g	150490	4452	414	2.0e-22	1.9e-20	qualified

Per-point arithmetic provenance

Table 3: Each of the 100 ordinary points per process evaluated once by a private clone with bridge route counters around every amplitude call (`tools/tirust_provenance.py`). MadLoop clears the TIRust and Fortran coefficient caches per point, so the deltas cover every reduction a point reused. Cache-off and a 1 MiB budget (insertion refusals) reproduce every coefficient, accuracy and status bit for bit.

Process	binary64 only	DD tier	MPFR tier	genuine QP	failed	status = campaign	cache off identical	1 MiB budget identical
EW u d̄ → W⁺	100	0	0	0	0	100/100	yes	yes
EW e⁺e⁻ → μ⁺μ⁻	100	0	0	0	0	100/100	yes	yes
EW e⁺e⁻ → t t̄ γ	100	0	0	0	0	100/100	yes	yes
EW e⁺e⁻ → μ⁺μ⁻ γ γ	40	58	1	1	0	100/100	yes	yes
loop-induced g g → H	100	0	0	0	0	100/100	yes	yes
loop-induced g g → H H	100	0	0	0	0	100/100	yes	yes
QCD+QED u ū → g γ	100	0	0	0	0	100/100	yes	yes
QCD u ū → Z	100	0	0	0	0	100/100	yes	yes
QCD g g → t t̄	100	0	0	0	0	100/100	yes	yes
QCD u ū → Z g g	87	13	0	0	0	100/100	yes	yes
QCD u ū → Z Z g g	14	71	15	0	0	100/100	yes	yes

Physical stress points

Table 4: Threshold, soft/collinear, small-Gram and complex-mass points (`tools/tirust_stress.py`). Reference: forced-QP CutTools and Ninja when they agree, otherwise stability-checked values of independent reducers that agree. TIRust may refuse; only an accepted value that disagrees would block (0 such cases). Refusals where every reducer fails are properties of the point. “Process STOP”: CutTools ends the program (CT692) on the ZZgg soft-collinear point; on the most forward CMS small-Gram point it never returns. Any lane falling back to CutTools therefore aborts or hangs there; the driver records the point and resumes with the next one. “TIRust QP” compares unchecked forced QP (`CTModeRun=4`) with the reference. “differs” occurs only where the genuine-QP interface refuses tensors (QP failure latch set; e.g. 48 rank-4 boxes at the H H small-Gram point that the DP-input path with its 512-bit tier evaluates correctly); automatic mode never accepts such results.

Process	point	reference	TIRust alone	chain 9|6|7|1	TIRust QP
CMS EW e⁺e⁻ → μ⁺μ⁻	12 ordinary	QP CutTools = QP Ninja ×12	accepted-agrees (219) ×12	accepted-agrees (219) ×12	agrees ×12
CMS EW e⁺e⁻ → μ⁺μ⁻	z-resonance	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
CMS EW e⁺e⁻ → μ⁺μ⁻	z-resonance	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
CMS EW e⁺e⁻ → μ⁺μ⁻	z-resonance	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
CMS EW e⁺e⁻ → μ⁺μ⁻	internal-ww-threshold	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
CMS EW e⁺e⁻ → μ⁺μ⁻	internal-ww-threshold	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
CMS EW e⁺e⁻ → μ⁺μ⁻	internal-ww-threshold	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
CMS EW e⁺e⁻ → μ⁺μ⁻	small-gram	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
CMS EW e⁺e⁻ → μ⁺μ⁻	small-gram	checked CutTools	refused (420)	rescued-by-1-agrees (329)	differs
CMS EW e⁺e⁻ → μ⁺μ⁻	small-gram	unavailable or disputed	refused (420)	unrescued (process hang) (–)	–
EW e⁺e⁻ → μ⁺μ⁻	small-gram	checked CutTools	refused (420)	rescued-by-1-agrees (329)	differs
EW e⁺e⁻ → t t̄ γ	threshold	checked results disagree	refused (420)	rescued-by-6-unreferenced (328)	–
EW e⁺e⁻ → t t̄ γ	soft-collinear	unavailable or disputed	refused (420)	unrescued (420)	–
EW e⁺e⁻ → t t̄ γ	small-gram	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
EW e⁺e⁻ → μ⁺μ⁻ γ γ	soft-collinear	unavailable or disputed	refused (420)	unrescued (420)	–
EW e⁺e⁻ → μ⁺μ⁻ γ γ	small-gram	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
loop-induced g g → H H	threshold	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees

Process	point	reference	TIRust alone	chain 9|6|7|1	TIRust QP
loop-induced g g → H H	small-gram	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	differs
QCD+QED u ū → g γ	small-gram	unavailable or disputed	refused (420)	unrescued (420)	–
QCD g g → t t̄	threshold	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
QCD g g → t t̄	small-gram	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
QCD u ū → Z g g	threshold	unavailable or disputed	refused (420)	unrescued (420)	–
QCD u ū → Z g g	soft-collinear	unavailable or disputed	refused (420)	unrescued (420)	–
QCD u ū → Z g g	small-gram	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
QCD u ū → Z Z g g	threshold	unavailable or disputed	refused (420)	unrescued (420)	–
QCD u ū → Z Z g g	soft-collinear	unavailable or disputed	refused (420)	unrescued (process STOP) (–)	–
QCD u ū → Z Z g g	small-gram	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees

Stability (Appendix C of arXiv:1604.01363)

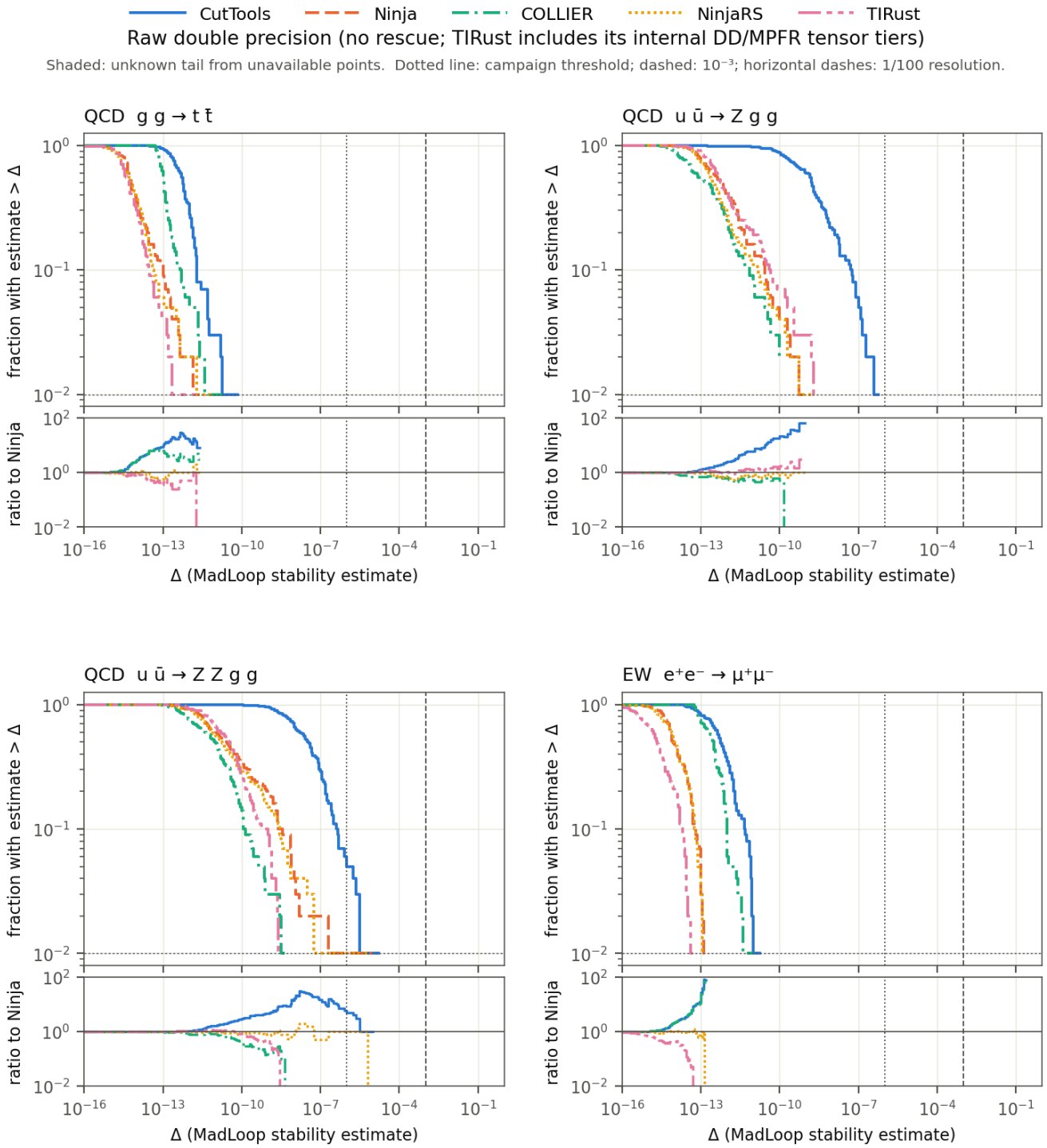


Figure 1: Raw double precision, no automatic rescue: MadLoop's internal estimate of the total answer on 100 frozen cut-qualified points per process. Shaded bands are unknown tails from unavailable points; a TIRust point is unavailable when its reduction refuses. TIRust's DP includes its internal DD/MPFR tensor tiers.

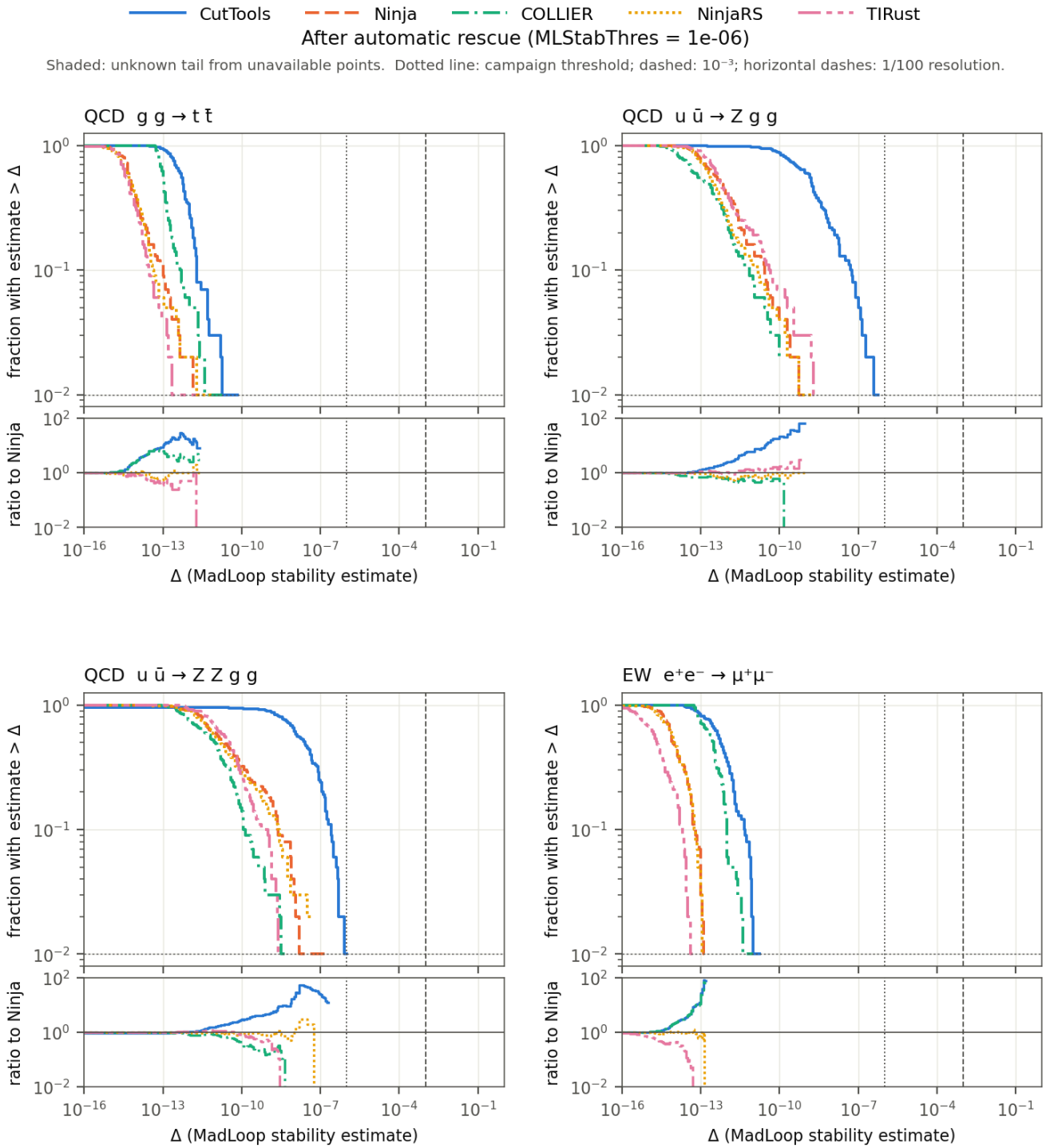


Figure 2: After MadLoop's automatic rescue at the campaign threshold `MLStabThres = 1e-6`.

Table 5: Fractions of the 100 points whose estimate exceeds the threshold; brackets give [known, known + unavailable] bounds. At 100 points the resolution is 0.01; larger samples remain a TODO. The estimate is MadLoop's self-consistency test, not an independent error.

Variant	Process	Reducer	available	> 1e-3	> 1e-6	> 1e-9
raw	QCD g g → t t̄	CutTools	100/100	0.00	0.00	0.00
raw	QCD g g → t t̄	Ninja	100/100	0.00	0.00	0.00
raw	QCD g g → t t̄	COLLIER	100/100	0.00	0.00	0.00
raw	QCD g g → t t̄	NinjaRS	100/100	0.00	0.00	0.00
raw	QCD g g → t t̄	TIRust	100/100	0.00	0.00	0.00
raw	QCD u ū → Z g g	CutTools	100/100	0.00	0.00	0.60
raw	QCD u ū → Z g g	Ninja	100/100	0.00	0.00	0.00
raw	QCD u ū → Z g g	COLLIER	100/100	0.00	0.00	0.00
raw	QCD u ū → Z g g	NinjaRS	100/100	0.00	0.00	0.01

Variant	Process	Reducer	available	> 1e-3	> 1e-6	> 1e-9
raw	QCD u ū → Z g g	TIRust	100/100	0.00	0.00	0.03
raw	QCD u ū → Z Z g g	CutTools	100/100	0.00	0.05	0.95
raw	QCD u ū → Z Z g g	Ninja	100/100	0.00	0.01	0.20
raw	QCD u ū → Z Z g g	COLLIER	100/100	0.00	0.00	0.03
raw	QCD u ū → Z Z g g	NinjaRS	100/100	0.00	0.01	0.15
raw	QCD u ū → Z Z g g	TIRust	100/100	0.00	0.00	0.09
raw	EW $e^+e^- \to \mu^+\mu^-$	CutTools	100/100	0.00	0.00	0.00
raw	EW $e^+e^- \to \mu^+\mu^-$	Ninja	100/100	0.00	0.00	0.00
raw	EW $e^+e^- \to \mu^+\mu^-$	COLLIER	100/100	0.00	0.00	0.00
raw	EW $e^+e^- \to \mu^+\mu^-$	NinjaRS	100/100	0.00	0.00	0.00
raw	EW $e^+e^- \to \mu^+\mu^-$	TIRust	100/100	0.00	0.00	0.00
rescued	QCD g g → t $\bar{t}$	CutTools	100/100	0.00	0.00	0.00
rescued	QCD g g → t $\bar{t}$	Ninja	100/100	0.00	0.00	0.00
rescued	QCD g g → t $\bar{t}$	COLLIER	100/100	0.00	0.00	0.00
rescued	QCD g g → t $\bar{t}$	NinjaRS	100/100	0.00	0.00	0.00
rescued	QCD g g → t $\bar{t}$	TIRust	100/100	0.00	0.00	0.00
rescued	QCD u ū → Z g g	CutTools	100/100	0.00	0.00	0.60
rescued	QCD u ū → Z g g	Ninja	100/100	0.00	0.00	0.00
rescued	QCD u ū → Z g g	COLLIER	100/100	0.00	0.00	0.00
rescued	QCD u ū → Z g g	NinjaRS	100/100	0.00	0.00	0.01
rescued	QCD u ū → Z g g	TIRust	100/100	0.00	0.00	0.03
rescued	QCD u ū → Z Z g g	CutTools	100/100	0.00	0.00	0.90
rescued	QCD u ū → Z Z g g	Ninja	100/100	0.00	0.00	0.19
rescued	QCD u ū → Z Z g g	COLLIER	100/100	0.00	0.00	0.03
rescued	QCD u ū → Z Z g g	NinjaRS	100/100	0.00	0.00	0.14
rescued	QCD u ū → Z Z g g	TIRust	100/100	0.00	0.00	0.09
rescued	EW $e^+e^- \to \mu^+\mu^-$	CutTools	100/100	0.00	0.00	0.00
rescued	EW $e^+e^- \to \mu^+\mu^-$	Ninja	100/100	0.00	0.00	0.00
rescued	EW $e^+e^- \to \mu^+\mu^-$	COLLIER	100/100	0.00	0.00	0.00
rescued	EW $e^+e^- \to \mu^+\mu^-$	NinjaRS	100/100	0.00	0.00	0.00
rescued	EW $e^+e^- \to \mu^+\mu^-$	TIRust	100/100	0.00	0.00	0.00

Provisional stock O3 profile (not acceptance)

Table 6: Stock `check profile` of QCD $u\bar{u} \to Zggg$ on the existing O3 export (learned filters, 48/96 helicities, both poles, timed `CTModeRun=1` without stability test), 50 points, one CPU, three repetitions in alternating reducer order, TIRust DSO `6c11cfd9eecf`. TIRust/NinjaRS 1.177 (the 48-page report measured 1.558 with the previous runtime). Unpaired shared-host data: no confidence bound, no acceptance claim.

Reducer	FULL ms/point (median)	three repetitions	initialization s	ratio / NinjaRS
CutTools	1028.5	1021.5 / 1040.9 / 1028.5	32.2	1.069
Ninja	968.2	970.2 / 968.2 / 967.2	12.8	1.007
COLLIER	1183.5	1183.5 / 1189.4 / 1176.3	73.5	1.230
NinjaRS	961.9	978.2 / 961.9 / 958.6	11.8	1.000
TIRust	1128.1	1151.1 / 1126.8 / 1128.1	8.0	1.173

Campaign `tirust-9a781998-1` (manifest `bb8b530e57a6`), exporter `be13df837e4f`. Commands, summaries and the local timing handoff are in `docs/benchmarkfolder/campaigns/tirust-9a781998-1/`. The preceding 48 pages are unchanged.